Course structure

- The Pearson Edexcel International Advanced Level in Chemistry comprises six units and contains an International Advanced Subsidiary subset of three IAS units.
- The International Advanced Subsidiary is the first half of the International Advanced Level course and consists of Units 1, 2 and 3. It may be awarded as a discrete qualification or contribute 50 per cent of the total International Advanced Level marks.
- The full International Advanced Level award consists of the three IAS units (Units 1, 2 and 3), plus three IA2 units (Units 4, 5 and 6) which make up the other 50 per cent of the International Advanced Level. Students wishing to take the full International Advanced Level must, therefore, complete all six units.
- The structure of this qualification allows teachers to construct a course of study that can be taught and assessed either as:
    - distinct modules of teaching and learning with related units of assessment taken at appropriate stages during the course; or
    - a linear course which is assessed in its entirety at the end.

1.1 Unit description

Chemical ideas

This unit provides opportunities for students to develop the basic chemical skills of formulae writing, equation writing and calculating chemical quantities.

The study of energetics in chemistry is of theoretical and practical importance. In this unit students learn to define, measure and calculate enthalpy changes. They will see how a study of enthalpy changes can help chemists to understand chemical bonding.

The study of atomic structure introduces s, p, and d orbitals and shows how a more detailed understanding of electron configurations can account for the arrangement of elements in the periodic table.

The unit introduces the three types of strong chemical bonding (ionic, covalent and metallic).

Organic chemistry is also introduced, with students studying alkanes and alkenes.

How chemists work

Practical work measuring energy changes helps students to understand the ideas of uncertainty in measurements and evaluate their results in terms of systematic and random errors.

The study of atomic structure gives some insight into the types of evidence which scientists use to study electrons in atoms. This leads to an appreciation of one of the central features of chemistry which is the explanation of the properties of elements and the patterns in the periodic table in terms of atomic structure.

The role of instrumentation in analytical chemistry is illustrated by mass spectrometry.

Students are introduced to some of the evidence which will help them to understand the different kinds of chemical bonding.

Chemists set up theoretical models and gain insights by comparing real and ideal properties of chemicals. This is illustrated in the unit by the ionic model and the comparison of lattice energies calculated from theory with those determined with the help of Born-Haber cycles.

Throughout the unit students see the importance of chemical data and learn to select data from databases and use it to look for patterns and calculate other quantities.

The introduction to organic chemistry shows how chemists work safely with potentially hazardous chemicals by managing risks.

Chemistry in action

The uses of mass spectrometry illustrate the importance of sensitive methods of analysis in areas such as space research, medical research and diagnosis, in detecting drugs in sport and in environmental monitoring.

In this unit students learn how chemical insights can help to make the use of polymeric and other materials more sustainable. This involves analysis of the uses of energy, raw materials and other resources at each stage of the life cycle of products.

Core practicals

The following specification points are core practicals within this unit that students should complete:

1.3j

1.3k

1.4f

These practicals may appear in the written examination for Unit 1.

Use of examples

Examples in practicals

Where 'e.g.' follows a type of experiment in the specification students are not expected to have carried out that specific experiment. However, they should be able to use data from that or similar experiments.

For instance in this unit, *1.4f ii Energetics*, the specification states:

simple enthalpy of combustion experiments using, e.g. a series of alcohols in a spirit burner.

Students should have carried out simple enthalpy of combustion reactions, but they may or may not have carried out these using alcohol in spirit burners.

In the unit test students could be given experimental data for this, or any other enthalpy of combustion reaction, and be expected to analyse and evaluate this data.

Examples in unit content

Where 'e.g.' follows a concept students are not expected to have been taught the particular example given in the specification. They should be able to illustrate their answer with an example of their choice.

For instance in this unit, *1.6.1f Ionic bonding*, the specification states:

recall trends in ionic radii down a group and for a set of isoelectronic ions, e.g. N^{3-} *to* Al^{3+}.

Students will be expected to recall the trends in ionic radii down a group, and for a set of isoelectronic ions, but they may or may not have done this from N^{3-} to Al^{3+}.

In the unit test students could be asked to recall the trends in ionic radii down a group. They could be asked this in reference to any group in the periodic table, either the one listed as an example or another group.